Improvements to the extension plan

Generalizing the two-core machinery to arbitrary d, partitions, and Ω :: August 28, 2026

The answer to your question is yes: the two arguments developed for the three-dimensional two-core case — the confinement (tilt) argument and the first-moment argument — generalize with no essential change to the full setting of the extension plan: arbitrary dimension, arbitrary nonempty partition $(\mathcal{C}, \mathcal{R})$, and arbitrary positive-definite correlation matrix Ω. This upgrades the plan in three places: Proposition 1's hypothesis can be weakened, a new general admissibility result becomes available for $|\mathcal{C}| \geq 2$, and the paper acquires a cleaner organizing principle — a dichotomy in $|\mathcal{C}|$ that explains, in one line, why the $\mathcal{C} = \{1\}$ and $|\mathcal{C}| \geq 2$ cases behave so differently. Notation follows the plan; throughout, A is closed, convex, coordinatewise lower, with $P_0(X \in A) = 1 - \alpha$, and $\mathcal{F}(\Omega, \Lambda) = \{a \geq 0 : \Omega a \in \Lambda\}$.

1 The organizing dichotomy

For a cone K, write $K^* := \{d : \mu' d \geq 0 \text{ for all } \mu \in K\}$ for its dual cone. In the two-core note, Λ_1^* was called the transfer cone. The following computation is elementary but, I think, should be a displayed lemma in the paper, because everything downstream is governed by it.

Lemma 1 (Transfer-cone dichotomy)**.** *(i) If $|\mathcal{C}| = 1$, say $\mathcal{C} = \{1\}$, then*

$$\Lambda_1^* = \Big\{ d \in \mathbf{R}^d : \ d_r \leq 0 \text{ for all } r \in \mathcal{R}, \ d_1 + \textstyle\sum_{r \in \mathcal{R}} d_r \geq 0 \Big\},$$

a nontrivial polyhedral cone containing the gate directions $e_1 - e_r$. (ii) If $|\mathcal{C}| \geq 2$, then $\Lambda_1^* = \{0\}$.

Proof. (i) Normalize $\mu_1 = 1$. The constraint $d_1 + \sum_r \mu_r d_r \geq 0$ for all $\mu_r \leq 1$ forces $d_r \leq 0$ (send $\mu_r \to -\infty$), and the infimum is then attained at $\mu_r = 1$, giving $d_1 + \sum_r d_r \geq 0$; conversely these two conditions suffice. (ii) Let $d \in \Lambda_1^*$. For distinct $c, c' \in \mathcal{C}$, the mean $e_c + s e_{c'}$ lies in Λ_1 for every $s \in \mathbf{R}$, so $d_c + s d_{c'} \geq 0$ for all s, forcing $d_{c'} = 0$; running over ordered pairs gives $d_c = 0$ for all $c \in \mathcal{C}$. For $r \in \mathcal{R}$, the mean with $\mu_c = 1$ for one core, $\mu_r = 1$, and all other redundant coordinates at $-M$ lies in Λ_1; letting $M \to \infty$ forces $d_{r'} \leq 0$ for every r', and setting all redundant coordinates equal to 1 gives $\sum_r d_r \geq 0$, whence $d_r = 0$ for all r. □

Two consequences frame the whole plan. First, when $|\mathcal{C}| = 1$ the set Λ is convex (the constraint $\mu_r \leq \mu_1$ is linear), so $\mathcal{F}(\Omega, \Lambda)$ is a convex cone; when $|\mathcal{C}| \geq 2$, Λ is a nonconvex union of branches and $\mathcal{F}$ is nonconvex. Second, the nonzero elements of Λ_1^* for $|\mathcal{C}| = 1$ are precisely the pointwise transfer directions that generate the gated dominators of Section 5 of the plan, while for $|\mathcal{C}| \geq 2$ no pointwise transfer exists and — as shown below — every weak dominator of an exact-size test must match its full first-moment vector under P_0. Domination constructions live exactly where $\Lambda_1^* \neq \{0\}$; admissibility results live where $\Lambda_1^* = \{0\}$. I would state this dichotomy early and hang Sections 3–5 of the plan on it.

2 Upgrade 1: weaken Proposition 1 to feasible support

Say A is *feasibly supported* if

$$A = \bigcap_{a \in F} \{x : a'x \le h_A(a)\}, \qquad F := \{a \in \mathcal{F}(\Omega, \Lambda) : a \neq 0,\ h_A(a) < \infty\},$$

i.e. the supporting halfspaces with feasible normals already determine A. For polyhedra this holds if and only if every facet normal is feasible. Proposition 1 of the plan assumes more: that *every* finite-support direction is feasible.

Proposition 1 (Feasible support suffices; general Ω)**.** *If A is feasibly supported, then ϕ_A is α-admissible against all level-α tests, including randomized competitors.*

Proof. Let ψ be level α and weakly dominate ϕ_A on Λ_1; set $h := \psi - \phi_A$, so $\{h > 0\} \subseteq A$ and $\{h < 0\} \subseteq A^c$. Coordinatewise lowerness and $\Lambda_0 \subseteq (-\infty, 0]^d$ make $\mu = 0$ least favorable, so $\mathrm{E}_0\psi \le \alpha = \mathrm{E}_0\phi_A$; dominance along the ray $te_c \in \Lambda_1$ ($c \in \mathcal{C}$) and continuity of power functions give the reverse inequality, hence $\mathrm{E}_0 h = 0$. Now fix $a \in F$. As in the plan's Proposition 1, $\Omega a \in \Lambda$ and $a'\Omega a > 0$ imply $\Omega a \in \Lambda_1$, and Λ_1 is a cone, so $t\,\Omega a \in \Lambda_1$ for all $t > 0$. Dominance along this mean ray and the canonical representation give $\mathrm{E}_0[e^{ta'X}h] \ge 0$ for all $t > 0$. Since h^+ is supported on $A \subseteq \{a'x \le h_A(a)\}$, the standard bound

$$\mathrm{E}_0[e^{ta'X}h] \le e^{th_A(a)}\mathrm{E}_0 h^+ - e^{t(h_A(a)+\varepsilon)}\mathrm{E}_0\big[h^-1\{a'X \ge h_A(a) + \varepsilon\}\big]$$

shows that $\mathrm{E}_0[h^-1\{a'X \ge h_A(a) + \varepsilon\}] = 0$ for every $\varepsilon > 0$, hence $h^- = 0$ a.s. on $\{a'x > h_A(a)\}$. The complement $A^c = \bigcup_{a \in F}\{a'x > h_A(a)\}$ is open, so by Lindelöf a countable subfamily covers it and $h^- = 0$ a.s. on A^c, hence a.s. Then $\mathrm{E}_0 h = 0$ forces $h = 0$ P_0-a.s., and mutual absolute continuity of the Gaussian shifts makes a strict improvement impossible. □

Three remarks. First, the proof is elementary and self-contained; Lehmann and Romano's Theorem 6.7.1 is no longer needed, though it can still be cited for context. Second, when $|\mathcal{C}| = 1$ the upgrade is vacuous: $\mathcal{F}(\Omega, \Lambda)$ is then a convex cone (Lemma 1), the finite-support directions of A form the closed convex cone generated by its facet normals, and feasible support is equivalent to the plan's condition. The strict gain occurs exactly when $|\mathcal{C}| \ge 2$, where $\mathcal{F}$ is nonconvex: sums of feasible facet normals may leave $\mathcal{F}$, and the plan's condition then fails for regions that Proposition 1 still covers (the symmetric gate of the three-dimensional note is the canonical example). So the weakening is free precisely where the current plan has the least to say. Third, Proposition 2 of the plan is an immediate corollary, unchanged.

3 Upgrade 2: a new result for $|\mathcal{C}| \ge 2$

The first-moment argument of the two-core note generalizes verbatim. Let ψ weakly dominate ϕ_A and $h = \psi - \phi_A$, so $\mathrm{E}_0 h = 0$ as above. For $\mu^* \in \Lambda_1$, dominance along the mean ray $t\mu^*$ and differentiation at $t = 0^+$ (justified since $|h| \le 1$ and the Gaussian m.g.f. is finite) give $(\Omega^{-1}\mu^*)'q \ge 0$ where $q := \mathrm{E}_0[Xh(X)]$, i.e. $\mu^{*\prime}(\Omega^{-1}q) \ge 0$ for every $\mu^* \in \Lambda_1$. Thus $\Omega^{-1}q \in \Lambda_1^*$, and Lemma 1 gives:

Lemma 2. *If $|\mathcal{C}| \geq 2$, every weak dominator of an exact-size test satisfies $q = \mathrm{E}_0[Xh] = 0$.*

Proposition 2 (Feasible support plus one halfspace; $|\mathcal{C}| \geq 2$, general Ω)**.** *Suppose $|\mathcal{C}| \geq 2$ and $A = A_g \cap \{x : u'x \leq \kappa\}$, where A_g is an intersection of halfspaces with normals in $\mathcal{F}(\Omega, \Lambda)$ and $u \neq 0$ is arbitrary. Then ϕ_A is α-admissible.*

Proof. The confinement step of Proposition 1 gives $h^- = 0$ a.s. on A_g^c, so h^- is supported on $\{u'x > \kappa\} \cap A_g$ while h^+ is supported on $A \subseteq \{u'x \leq \kappa\}$. With $M := \mathrm{E}_0 h^+ = \mathrm{E}_0 h^-$, if $M > 0$ then $\mathrm{E}_0[u'Xh^+] \leq \kappa M$ and $\mathrm{E}_0[u'Xh^-] > \kappa M$ (strictly, since $u'X > \kappa$ a.s. on $\{h^- > 0\}$ and $P_0(u'X = \kappa) = 0$), so $u'q < 0$, contradicting Lemma 2. Hence $M = 0$ and the proof concludes as before. □

Its corollaries are the payoff.

Corollary 1 (Rectangles with one redundant moment)**.** *Let $|\mathcal{C}| \geq 2$, $|\mathcal{R}| = 1$, and Ω be any positive-definite correlation matrix. Every finite rectangle $A_{\boldsymbol{\kappa}} = \bigcap_j \{x_j \leq \kappa_j\}$ with $P_0(A_{\boldsymbol{\kappa}}) = 1 - \alpha$ is α-admissible.*

Proof. Each core face is feasible: $(\Omega e_c)_r = \Omega_{rc} < 1 = (\Omega e_c)_c \leq \max_{c' \in \mathcal{C}} (\Omega e_c)_{c'}$, since Ω is a nondegenerate correlation matrix. The single redundant face $\{x_r \leq \kappa_r\}$ is the one extra halfspace (it is never feasible: $(\Omega e_r)_r = 1 > \max_c \Omega_{cr}$). Apply Proposition 2. □

This generalizes the three-dimensional admissibility result for rectangles from $(d, \Omega) = (3, I_3)$ to arbitrary dimension and arbitrary correlation, in one stroke. Note the contrast with the plan's Section 5: with $|\mathcal{C}| = 1$ the same rectangles are strictly dominated. The rectangle's status flips exactly at the dichotomy of Lemma 1, which is, I think, the cleanest single sentence the paper can offer about identification versus inference.

Corollary 2 (All one-sided tests; $|\mathcal{C}| \geq 2$)**.** *Let $|\mathcal{C}| \geq 2$ and $a \geq 0$, $a \neq 0$, be* any *nonnegative weight vector — feasibility not required, including weights concentrated entirely on $\mathcal{R}$. Then $\phi(x) = I\{a'x > z_{1-\alpha}\sqrt{a'\Omega a}\}$ is α-admissible for every $\alpha \in (0, 1)$.*

Proof. Take $A_g = \mathbf{R}^d$ and $u = a$ in Proposition 2; the acceptance region is lower and has exact size at the origin. □

Corollary 2 strengthens the plan's Proposition 2 in the $|\mathcal{C}| \geq 2$ case in two directions: the perturbation parameter ε and the feasibility requirement $a_\varepsilon \in \mathcal{F}(\Omega, \Lambda)$ are unnecessary — any nonnegative weighting of core and redundant moments is admissible, including the redundant-only test — and no restriction on α is needed. The ε-construction remains exactly the right tool for $|\mathcal{C}| = 1$, where Lemma 2 fails and feasibility is doing real work. I would present Proposition 2 as the $|\mathcal{C}| = 1$ result and Corollary 2 as its $|\mathcal{C}| \geq 2$ counterpart.

4 Suggested edits to the plan, section by section

Section 3. Replace the hypothesis of Proposition 1 by feasible support (Proposition 1); the current version survives as the special case in which all finite-support directions are feasible,

and the equivalence of the two conditions when $|\mathcal{C}| = 1$ is worth a remark. The proof becomes self-contained.

Section 4. Keep Proposition 2 for $|\mathcal{C}| = 1$; add Corollary 2 for $|\mathcal{C}| \geq 2$. The paper's headline — redundant moments need not be irrelevant for inference — becomes stronger: with at least two core moments, *every* nonnegative weighting is admissible.

Section 5. Unchanged, but preface it with Lemma 1: the gate directions used in Propositions 3 and 4 are exactly the nonzero elements of Λ_1^* (for Proposition 4, of $\widetilde{\Lambda}_1^*$ — condition (16) is precisely what makes that dual cone nontrivial). This unifies the two halves of the paper: dominators exist when the relevant dual cone is nontrivial, and when it is trivial the first-moment identity $q = 0$ converts the same tilt comparisons into admissibility proofs.

Section 6, step 4. The claim that the feasible-normal condition is "both sufficient and necessary" should be scoped to the bivariate ($|\mathcal{C}| = 1$) model, where $\mathcal{F}$ is convex and the gated dominators supply necessity. For $|\mathcal{C}| \geq 2$ necessity fails badly — Corollaries 1 and 2 are admissible tests violating it — and the correct statement of what remains open is: nice regions requiring at least two infeasible supporting halfspaces (e.g. rectangles with $|\mathcal{C}| \geq 2$ and $|\mathcal{R}| \geq 2$, or a rectangle plus a binding infeasible cut). Whether any such test is inadmissible is exactly the two-core frontier question discussed in my previous note, now posed at full generality; my conjecture there (all nice tests admissible when $\Lambda_1^* = \{0\}$) extends to this setting with the same caveats.

One further remark on scope: everything above uses only that Λ_1 is a cone containing the directions listed in the proof of Lemma 1, that $\Lambda_0 \subseteq (-\infty, 0]^d$, and that A is lower with exact size. So the machinery is robust to the refinements of Λ contemplated in Section 5.2 of the plan: for a maintained $\widetilde{\Lambda} \subseteq \Lambda$, the two regimes are governed by $\widetilde{\Lambda}_1^*$, and Propositions 1–2 hold with $\mathcal{F}(\Omega, \widetilde{\Lambda})$ and $\widetilde{\Lambda}_1^* = \{0\}$ in the obvious places.